BEING SMOOTH IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Characterisation of smooth morphisms by local finite presentation, flatness and smooth fibres. Morphisms, the smooth chapter's characterisation, quoted at proof lines 6393-6394. Splits smoothness into three properties.
- Flatness and local finite presentation are fpqc local on the base. Same chapter, Lemmas lemma-descending-property-flat and lemma-descending-property-locally-finite-presentation, invoked at proof lines 6394-6398. Gives two of the three ingredients.
- The set of points where the fibres are smooth is fpqc local on the base. Morphisms, Lemma morphisms-lemma-set-points-where-fibres-smooth, invoked at proof lines 6399-6400. Gives the third ingredient and hence the result.

Source reference. Original lemma 02L2: sources/stacks-descent/pinned-original-source.tex, source lines 6050–6050. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KY: sources/stacks-descent/pinned-original-source.tex, source lines 5984–5984. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02V4: sources/stacks-chow/tags.txt, source lines 3755–3755. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$ "f is smooth" is fpqc local on the base.*

Proof. A morphism is smooth if and only if it is locally of finite presentation, flat, and has smooth fibres. We have seen already that being flat and locally of finite

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presentation are fpqc local on the base (Lemmas 02L2, and 02KY). Hence the result follows for smooth from Morphisms, Lemma 02V4. □